

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM agents to autonomously decide when to act directly, self-reflect, or consult external knowledge during planning tasks.

arxiv.org/abs/2504.03553

The Problem: Indiscriminate "Flood Irrigation" in Agent Training

1. Trajectory Imitation

e.g., ReAct

Blindly imitates gold trajectories

2. Trial-and-Error

e.g., Reflexion

Indiscriminate external feedback injection

3. Knowledge-Augmented

e.g., KnowAgent, WKM

100% knowledge at every step regardless of need

Consequences of "Flood Irrigation"

- Fragile planning — pattern collapse on unexpected inputs
- Unnecessarily high inference costs (100% knowledge retrieval)
- No situational awareness — agents can't judge their own competence

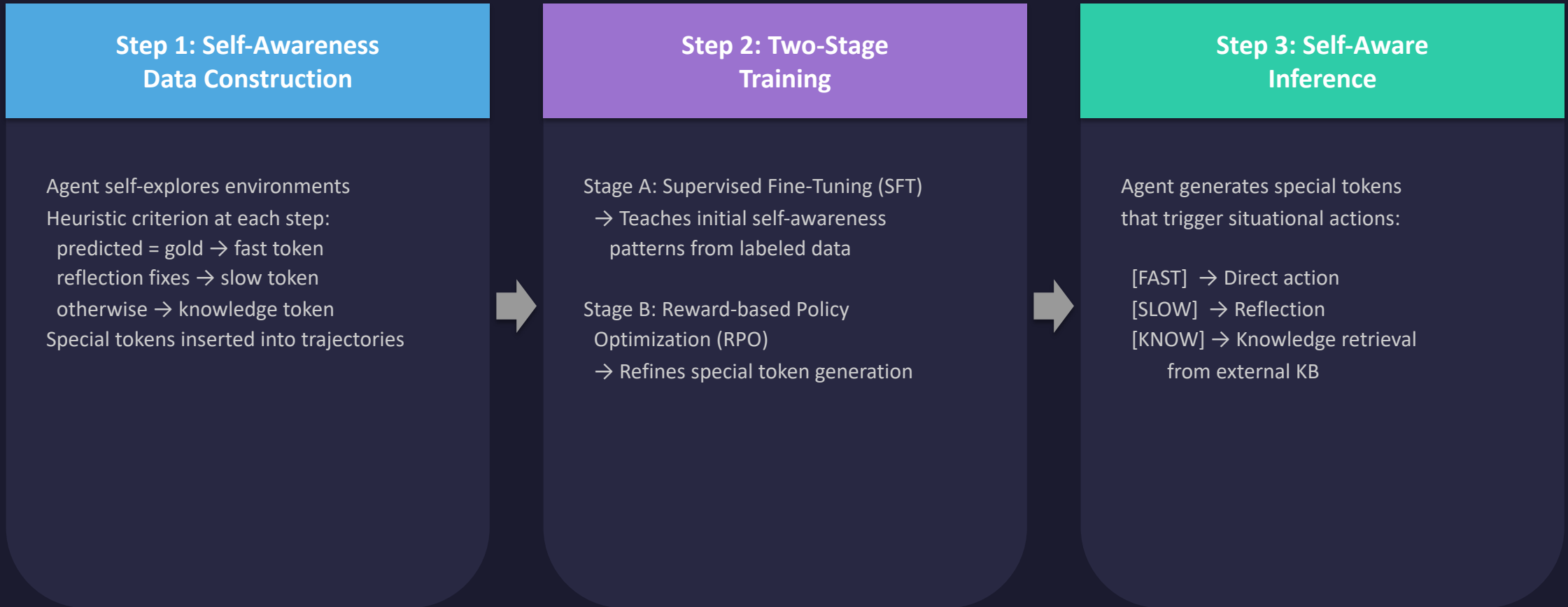
Human Contrast: Situational self-awareness — metacognitive ability to assess whether to act directly, pause and reflect, or seek external help. KnowSelf brings this to LLM agents.

Three Situational Thinking Modes



Agents autonomously regulate cognitive resources — analogous to human metacognition (Kahneman's System 1 / System 2 + external knowledge)

KnowSelf Pipeline: Data Construction → Training → Inference



Lightweight & data-efficient: requires only heuristic labeling on self-explored trajectories — no expensive human annotation or massive external feedback.

Qualitative Examples: Where Baselines Fail, KnowSelf Succeeds

Task A: "Put a hot mug in cabinet"

OpenAI-O1:

After opening microwave, incorrectly tries to take an egg from the microwave.

✗ Fails — wrong object, wrong action

KnowSelf (Llama-8B):

Uses [REFLECTION] token to realize it should heat the mug using the open microwave.

✓ Succeeds — self-corrects via slow thinking

Task B: "Put a saltshaker in drawer"

DeepSeek-R1:

Reflects that the drawer is open, then tries to close it first — which fails.

✗ Fails — incorrect sub-goal from reflection

KnowSelf (Llama-8B):

Applies [KNOWLEDGE] token: "place the object without removing unrelated items."

✓ Succeeds — retrieves targeted knowledge

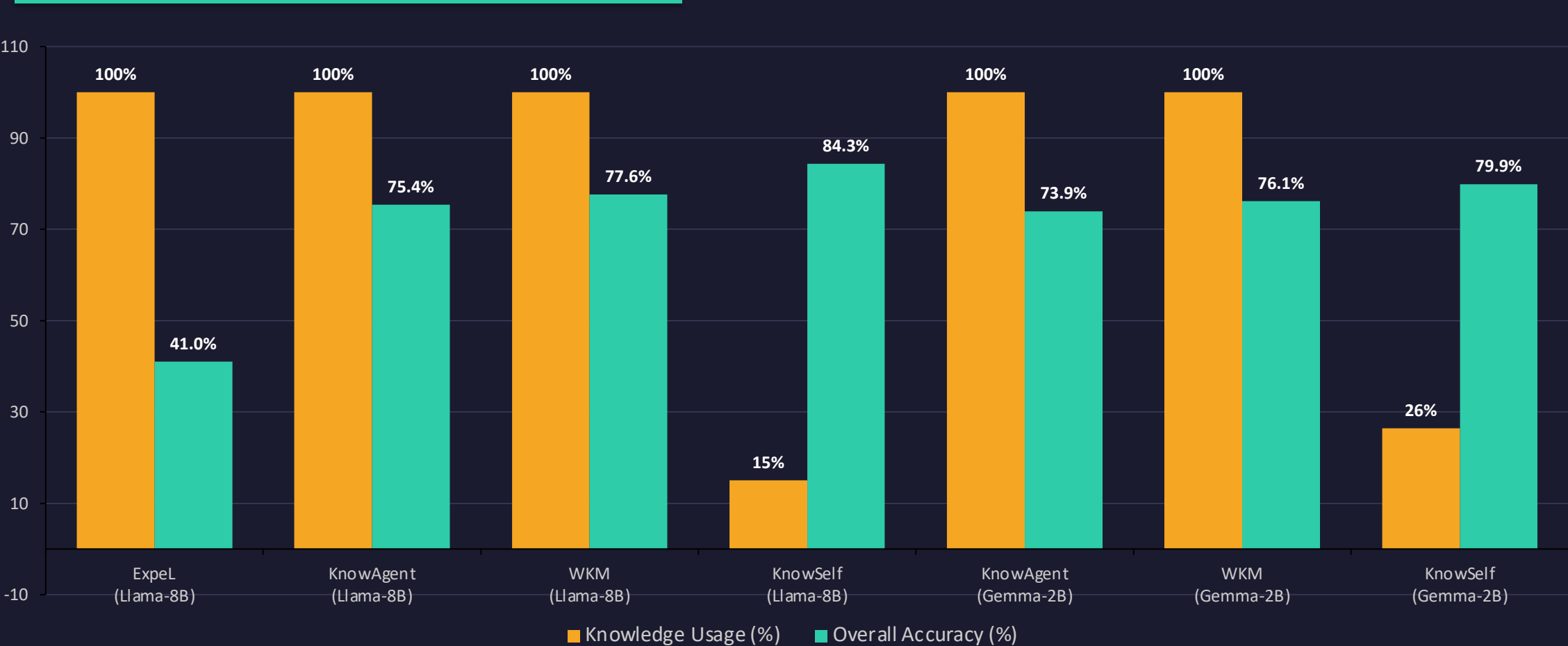
KnowSelf selects the right cognitive mode per situation — reflection when self-correction suffices, knowledge when it doesn't.

ALFWorld Results: KnowSelf Achieves 84.33% with Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Success rates (%) on six ALFWorld task types. Know% = proportion of steps using external knowledge.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Edge



Llama-8B KnowSelf: 84.33% accuracy with 15% knowledge
vs. WKM: 77.61% with 100% knowledge (+6.72 pts)

Gemma-2B KnowSelf: 79.85% accuracy with 26% knowledge
Matches GPT-4o ExpeL (79.85%) with a 2B model

Key Takeaways & Future Directions

1

Indiscriminate knowledge injection is wasteful and suboptimal

Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf

2

Agents can learn situational self-awareness via special tokens

Lightweight heuristic labeling on self-explored trajectories — no expensive annotation

3

KnowSelf (Llama-8B) outperforms GPT-4o baselines

84.33% vs. GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%)

4

Only 15–26% knowledge usage vs. 100% for competing methods

Massive efficiency gain without sacrificing — in fact improving — accuracy

5

The paradigm scales across model sizes

Even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL performance

Future Directions

- Extend to multi-agent collaborative planning with shared metacognition
- Apply the three-situation taxonomy to broader task domains (web, code, robotics)
- Explore learned (vs. heuristic) situation classifiers for data construction
- Investigate curriculum learning over thinking-mode distributions
- Study interaction between model scale and optimal knowledge utilization ratio